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# Recombinant Apolipoprotein A-$I_{Milano}$ Dimer Inhibits Carotid Intimal Thickening Induced by Perivascular Manipulation in Rabbits

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***Abstract*** Apolipoprotein A-$I_{Milano}$ (apoA-$I_M$), a natural variant of apolipoprotein A-I (apoA-I), confers to the carriers a significant protection against vascular disease. The antiatherogenic activity of a recombinant disulfide-linked apoA-$I_M$ dimer (rA-$I_M$/A-$I_M$) was analyzed in vivo by evaluating its effect on neointimal formation induced by periarterial manipulation in 1% cholesterol–fed rabbits. A flexible collar was applied around the carotid artery 21 days after the beginning of the dietary regimen, and animals were killed 10 days later. Rabbits were injected five times with reconstituted high-density lipoprotein containing egg phosphatidylcholine (EPC) and rA-$I_M$/A-$I_M$ (119 mg EPC+40 mg protein per dose) or with EPC liposomes (119 mg EPC per dose) beginning either 5 days before or at the day of collar positioning. Neither treatment affected plasma cholesterol levels. A significant intimal thickening was observed in control animals; the intima-to-media (I/M) ratio was 0.63±0.11 versus 0.03±0.05 for the sham-operated contralateral arteries. Neointimal formation was markedly inhibited in animals pretreated with rA-$I_M$/A-$I_M$ before lesion induction (I/M, 0.26±0.19) but not in those in which treatment began the day of collar insertion (I/M, 0.74±0.14). EPC liposomes did not affect neointimal formation (I/M, 0.50±0.14 and 0.51±0.07 in the two treatment groups). Proliferation of smooth muscle cells, assessed by direct incorporation of bromo-2′-deoxyuridine (BrdU) into replicating DNA, was reduced by ≈30% and 75% in the intimal and medial tissues of rA-$I_M$/A-$I_M$–pretreated rabbits. Thus, a short-term treatment with rA-$I_M$/A-$I_M$ inhibits smooth muscle cell proliferation and intimal thickening in rabbits, providing evidence for the potential use of rA-$I_M$/A-$I_M$ in the treatment of atherosclerosis. (***Circ Res.* 1995;76:405-411.**)

***Key Words*** • recombinant apolipoproteins • smooth muscle cell proliferation • atherosclerosis • restenosis

High-density lipoproteins (HDLs) play a major role in the so-called reverse cholesterol transport,[1] the process by which excess cholesterol in peripheral tissues, including the arterial wall, is transported to the liver for elimination; this possibility explains the strong inverse correlation between plasma HDL levels and coronary heart disease in many prospective and case-control studies.[2] The beneficial effects of elevated plasma HDL concentrations against arterial lipid deposition have suggested testing the activity of exogenous HDL on arterial lesions in experimental animals. Badimon et al[3] demonstrated that four weekly injections of a plasma-derived HDL fraction into cholesterol-fed rabbits inhibit the formation of new lesions and may cause a regression of established atherosclerotic lesions.

Apolipoprotein A-I (apoA-I), the primary protein component of HDL, seems to be responsible for the antiatherogenic activity of HDL. Indeed, apoA-I participates in reverse cholesterol transport, being a cofactor in lecithin/cholesterol acyltransferase activation[4]; it is also a ligand for the putative HDL receptor[5] and stimulates cholesterol efflux from lipid-loaded cells.[6] It also displays peculiar properties not directly related to its major activities in lipoprotein metabolism, ie, stimulation of prostacyclin release from the arterial wall and its stabilization,[7,8] activation of fibrinolysis,[9] and modulation of complement function.[10] Because of all of these activities, apoA-I overexpression in atherosclerosis-susceptible mice converts them into atherosclerosis-resistant animals,[11] whereas apoA-I infusion in cholesterol-fed rabbits exerts an inhibitory effect on the progression of atherosclerotic lesions.[12] Taken together, these findings argue for a potential therapeutic use of apoA-I in the treatment of human atherosclerosis.[13]

Apolipoprotein A-$I_{Milano}$ (apoA-$I_M$), the first described molecular variant of human apolipoproteins,[14,15] seems to confer to the carriers a significant protection against vascular disease.[16] ApoA-$I_M$ is characterized by a cysteine for arginine substitution at position 173 in the primary sequence,[17] allowing the formation of disulfide-linked homodimers (A-$I_M$/A-$I_M$) and heterodimers with apoA-II. The presence of these covalently-linked apoA-I molecules is likely to be responsible for the increased metabolic stability of the carriers' HDL. Indeed, in an in vitro system, the interconversion of A-$I_M$ HDL is impaired,[18] whereas in vivo the A-$I_M$ dimers are removed from plasma at a slower rate than monomeric A-$I_M$ or normal A-I.[19] The A-$I_M$/A-$I_M$ has been produced in *Escherichia coli* in sufficient amounts for in vitro and in vivo studies; the recombinant protein (rA-$I_M$/A-$I_M$) proved to be structurally and functionally identical to the natural protein.[20] In the search of innovative approaches for the treatment of neointimal formation,[21] we evaluated the effects of a short-term treatment with rA-$I_M$/A-$I_M$ in cholesterol-fed rabbits, in which intimal

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thickening was induced by inserting a flexible extra-arterial collar around the common carotid artery.[22]

## Materials and Methods

### Preparation of Reconstituted HDL

rA-$I_M$/A-$I_M$ was a generous gift of Pharmacia AB. ApoA-I was isolated from human plasma as previously described.[17] Reconstituted HDL (rHDL), containing rA-$I_M$/A-$I_M$ or apoA-I, and egg phosphatidylcholine (EPC) were prepared by the cholate dialysis technique,[23] by using a ratio of 2.5:1 (wt/wt) in the starting mixture. Each preparation was characterized by nondenaturing polyacrylamide gradient gel electrophoresis,[24] protein cross-linking with dimethylsuberimidate,[25] and determination of the phospholipid/protein composition. Different homogeneous preparations of rA-$I_M$/A-$I_M$–containing rHDL were pooled, sterilized, and stored at 4°C until use. EPC liposomes were prepared by the same procedure, omitting rA-$I_M$/A-$I_M$ in the starting mixture.

### Surgical Procedures

Male New Zealand White rabbits (1.8 to 2 kg) (Charles River, Italy) were used for the study. Insertion of the flexible extra-arterial collar was performed essentially as described previously.[22,26] Rabbits were anesthetized by an intramuscular injection of xylazine (5 mg/kg) and ketamine (35 mg/kg). Animals were then placed in dorsal recumbency, a midline neck incision was made, and both carotid arteries were surgically exposed. A nonocclusive, biologically inert, soft, hollow Silastic collar was positioned around both carotid arteries. The collar was 1.5 cm long and touched the arterial circumference at two points, 1.0 cm apart. In each animal, the collar was removed from one carotid (sham-operated artery), and the wounds were sutured.

### Study Protocol

A short-term protocol treatment (10 days) was chosen to avoid protein-antibody induction: rabbits were intravenously injected with either reconstituted HDL (EPC+rA-$I_M$/A-$I_M$) or EPC liposomes. Twenty-five rabbits were fed a cholesterol-rich (1%) diet at daily amounts of 100 g for 31 days. Collar placement was performed 21 days after the beginning of the diet regimen, and all animals were killed 10 days later. Animals were randomly assigned to the following five treatment groups: group 1, control (hypercholesterolemic diet only); groups 2 and 4, rabbits receiving EPC liposomes or reconstituted HDL, respectively, at days 16, 18, 20, 22, and 24; and groups 3 and 5, rabbits receiving EPC liposomes or reconstituted HDL, respectively, at days 21, 23, 25, 27, and 29.

### Histological and Immunofluorescence Analysis

#### *Histological Analysis*

Histological analysis to assess neointimal formation was performed on carotid arteries from all animals (sham-operated and collar-applied carotid arteries). Two rabbits from each group were injected into the marginal ear vein with 5-bromo-2′-deoxyuridine (BrdU, 40 mg/kg) 3 hours before death. All animals were given a lethal dose of sodium pentobarbital, and the vasculature was perfused with 0.1 mol/L phosphate-buffered saline (PBS) for 10 minutes. The three animals from each group not injected with BrdU were perfusion-fixed for another 20 minutes with 3% glutaraldehyde buffered with 0.1 mol/L PBS at a pressure of 100 mm Hg. The carotids from BrdU-injected animals (20 carotids, two animals from each group) were fixed by immersion in absolute ethanol overnight.

Tissues were paraffin-embedded and stained with hematoxylin and eosin.[26] At least 600 cross sections (5 μm thick) were cut for each artery. Neointimal formation was measured on carotids from all animals (50 carotids from 25 animals) by an image-analysis system (see below).

#### *Immunofluorescence Analysis*

Immunofluorescence analysis was performed on ethanol-fixed carotid arteries. Serial sections (n=10) were used to detect BrdU, α-smooth muscle (α-SM) actin, or macrophages by indirect immunofluorescence with anti-BrdU (Becton Dickinson), anti–α-SM1 actin (Sigma Chemical Co), or RAM11 macrophage-specific (DAKO A/S) monoclonal antibodies, respectively. Staining with these antibodies was performed according to standard procedures using a fluorescein isothiocyanate–conjugated (FITC) anti-mouse IgG antibody for visualization (Sigma).[26] For the proliferation analysis, after BrdU-positive cell counting, all nuclei were stained with propidium iodide, and the labeling index was calculated as the percentage of BrdU-positive versus total nuclei.

The discrimination between proliferative smooth muscle cells or macrophages was obtained with a double-labeling procedure. Proliferative smooth muscle cells were identified by the double reaction with tetramethylrhodamine isothiocyanate (TRITC)-labeled anti-BrdU and FITC-labeled anti–α-SM1 actin. Proliferative macrophages were identified by the double reaction with TRITC-labeled anti-BrdU and FITC-labeled anti-RAM 11.

### Image-Analysis System

Total and BrdU-positive cells and the cross-sectional thickness of the intima and underlying media were analyzed by an image-analysis system interfaced with a Zeiss Axioscope microscope. Neointimal formation was quantified as the ratio of the cross-sectional thickness of intimal and medial tissue. Intima-to-media (I/M) ratios were calculated to normalize the data. The I/M value obtained from each animal is the mean of at least 1000 measurements, performed on at least 10 different sections. Thickness, as well as labeling indexes, were determined for each wall compartment (media and intima) for each carotid artery.[26] Incorporation of BrdU in replicating cells was monitored by evaluating the femoral bone marrow labeling[26] in each animal. The image-analysis system consisted of a Macintosh IIx computer (Apple) equipped with a Frame Grabber Card (QUICKCAPTURE, Data Translation), a Sony high-resolution video camera, and a Trinitron SuperMac 21-inch color monitor. All measurements were performed by using National Institutes of Health (NIH) software (IMAGE, version 1.44; Dr Wayne Rasband, NIH, Bethesda, Md).

### Statistics

Statistical significance of differences between groups of animals was determined by unpaired Student's *t* test. A value of $P<.05$ was considered to be significant.

## Results

All rA-$I_M$/A-$I_M$ was incorporated into stable lipid-protein complexes with no lipid-free apolipoprotein left in the final preparation. When analyzed by nondenaturing polyacrylamide gradient gel electrophoresis, the different preparations of rA-$I_M$/A-$I_M$–containing rHDL consistently displayed two major components with diameters of 8.6 and 12.9 nm, together with additional minor components (Fig 1). The small rHDL contained a single molecule of rA-$I_M$/A-$I_M$, as assessed by cross-linking with dimethylsuberimidate[25]; two rA-$I_M$/A-$I_M$ molecules were present in the large rHDL. This particle size distribution is clearly different from that obtained with normal apoA-I, characterized by a single major population of rHDL, with a diameter of 9.3 nm and several minor components (Fig 1). The formation of heterogeneous rHDL with rA-$I_M$/A-$I_M$ is mostly due to the disulfide linkage in rA-$I_M$/A-$I_M$, which drastically modifies the physicochemical properties of apoA-I[20] and, in

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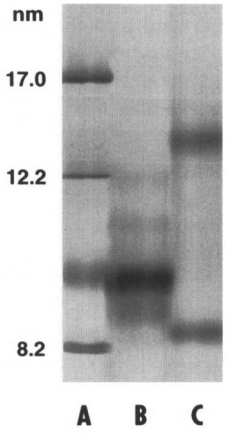



FIG 1. Nondenaturing polyacrylamide gradient gel electrophoresis of reconstituted high-density lipoprotein (rHDL) containing egg phosphatidylcholine and apolipoprotein A-I (apoA-I) or recombinant disulfide-linked apoA-$I_{Milano}$ (rA-$I_M$/A-$I_M$). Lanes are as follows: A, protein standards; B, apoA-I–containing rHDL; and C, rA-$I_M$/A-$I_M$–containing rHDL.

turn, its lipid-binding behavior. Indeed, heterogeneous $HDL_3$ subpopulations have been identified in the plasma of A-$I_M$ carriers, whereas a single population of $HDL_3$ is present in control plasma.[25] The protein concentration of the final rHDL preparation was 3.7 mg/mL with an EPC/protein ratio (wt/wt) of 2.98. Each dose contained 40 mg protein and 119 mg of EPC in a final volume of 10.8 mL; the cholate concentration was <0.5 mmol/L.

Plasma cholesterol levels in control rabbits (group 1) increased progressively from 40 to 60 to 930±245 mg/dL at the day of collar insertion and to 1741±144 mg/dL at the end of the study. These values did not differ from those of animals receiving EPC liposomes (groups 2 and 3) (909±175 mg/dL at surgery and 1875±320 mg/dL at the time of death) or rA-$I_M$/A-$I_M$ (groups 4 and 5) (941±225 mg/dL at surgery and 1923±484 mg/dL at the time of death). Agarose gel electrophoresis of serum samples demonstrated that the hypercholesterolemia in all rabbits was accompanied by the accumulation of atherogenic β-very-low-density lipoprotein particles.

The contralateral carotid arteries showed normal histology in all treated groups (Fig 2, top left), with no thickening of the intimal or medial tissue (Table), as previously described.[26] This applied to all rabbits independently of the treatment. Periarterial insertion of the flexible collar in control animals (group 1) resulted in a marked increase in intimal thickness (mostly cellular) in the carotid arteries with the collar (Fig 2, top right); medial thickness did not change significantly. The extent of intimal thickening induced by the collar in control animals (I/M ratio of 0.63±0.11) was within the range observed in a larger series of cholesterol-fed rabbits (0.63±0.07, n=10). Intimal carotid thickening in EPC-treated rabbits (groups 2 and 3) (Fig 2, bottom left) was slightly lower than in control rabbits (Table), possibly confirming previous reports describing modest antiatherogenic activity of phospholipid infusions in experimental animals.[27] Treatment with rA-$I_M$/A-$I_M$ significantly inhibited the formation of intimal lesions after perivascular manipulation only when it was started before collar insertion (group 4) (Fig 2, bottom right). The mean intimal thickness of the rA-$I_M$/A-$I_M$–pretreated rabbits was ≈40% and ≈50% of that in control and EPC-treated rabbits, respectively (Table). The intima of rA-$I_M$/A-$I_M$–pretreated rabbits contained significantly fewer cell layers than the intima of control and EPC-

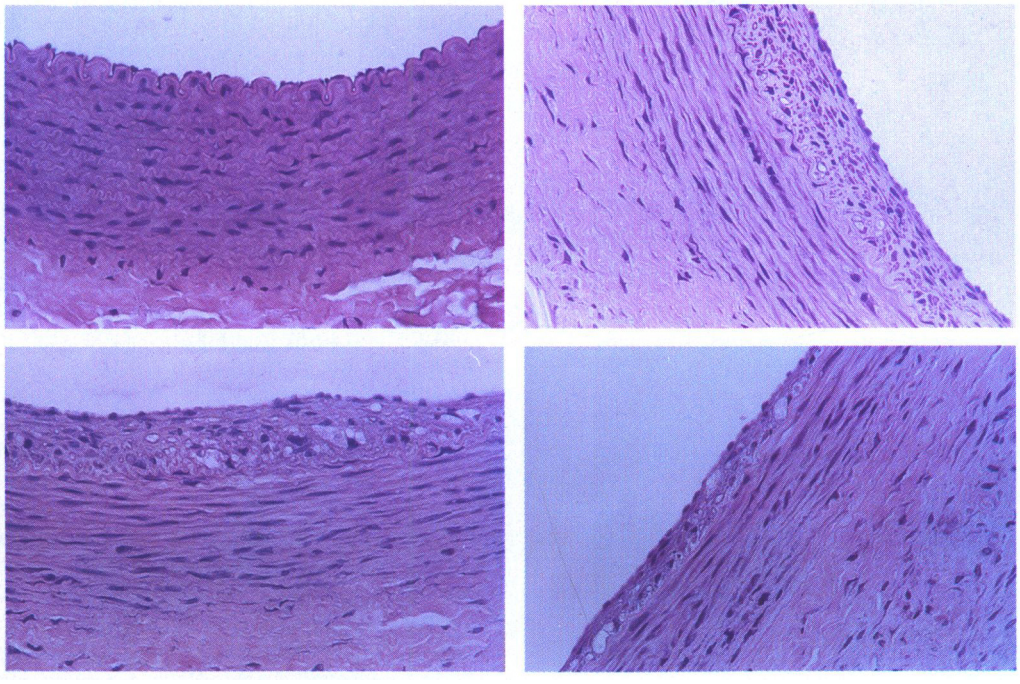


FIG 2. Microscopic views of transverse sections (original magnification ×400) (5 μm thick) through the carotid arteries of hypercholesterolemic rabbits 10 days after collar positioning. Top left, Sham-operated artery without intimal thickening. Top right, Artery with collar from a control rabbit showing marked neointimal hyperplasia. Bottom left, Artery with collar from an egg phosphatidylcholine–treated rabbit showing neointimal hyperplasia similar to control. Bottom right, Artery with collar from a recombinant disulfide-linked apolipoprotein A-$I_{Milano}$–pretreated rabbit showing a marked reduction in neointimal formation.

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**Intimal Thickening and 5-Bromo-2′-deoxyuridine Labeling of Intimal and Medial Tissues of Carotid Arteries With and Without Collar in Control Rabbits and in Rabbits Treated With Egg Phosphatidylcholine Liposomes or Recombinant Disulfide-Linked Apolipoprotein A-$I_{Milano}$ Dimer–Containing Reconstituted High-Density Lipoprotein**

| Group | Treatment | No. of Arteries | Thickness, μm | | Intima/Media Ratio | Labeling Index | |
|---|---|---|---|---|---|---|---|
| | | | Intima | Media | | Intima | Media |
| 1 | Control | 5 | 68.0±27.0 | 116±18 | 0.63±0.11 | 6.1±2.1 | 4.4±1.4 |
| 2 | EPC | 5 | 61.0±28.0 | 127±26 | 0.50±0.14 | 5.3±3.3 | 3.5±2.4 |
| 3 | EPC | 5 | 87±9.0 | 172±30 | 0.51±0.07 | 5.2±2.0 | 5.5±0.4 |
| 4 | rA-$I_M$/A-$I_M$ | 5 | 29.0±23.0 | 109±22 | 0.26±0.19* | 4.2±1.0* | 1.2±1* |
| 5 | rA-$I_M$/A-$I_M$ | 5 | 98±26 | 128±19 | 0.74±0.14 | 5.8±1.5 | 6.3±4.1 |
| Sham | All | 25 | 5.0±0.7 | 157±38 | 0.03±0.05 | ND | 0.01±0.02 |

EPC indicates egg phosphatidylcholine; rA-$I_M$/A-$I_M$, recombinant apolipoprotein A-$I_{Milano}$ dimer; sham, no collar applied; and ND, not determined. Values from serial cross sections are expressed as mean±SD. In groups 2 and 4, treatment began 5 days before surgery; in groups 3 and 5, treatment began the day of collar positioning.
*$P<.05$ vs control or EPC.

treated rabbits. By contrast, when rA-$I_M$/A-$I_M$ was administered at the day of collar insertion and thereafter, no significant reductions in intimal thickening were observed (Table).

Cell-specific antibodies were used to evaluate the composition of intimal and medial cell populations by immunostaining techniques. The medial tissues of sham-operated arteries and collar-applied arteries of all animals showed a strong reactivity for α-SM actin. The neointima induced by collar positioning contained predominantly α-SM actin–positive cells (>90% of the whole cell population) (Fig 3, top). The neointima but not the media also contained a few macrophages, identified by a positive reaction with a specific antibody, RAM11 (Fig 3, bottom). Thus, the lesions induced by collar insertion consisted mainly of smooth muscle cells.

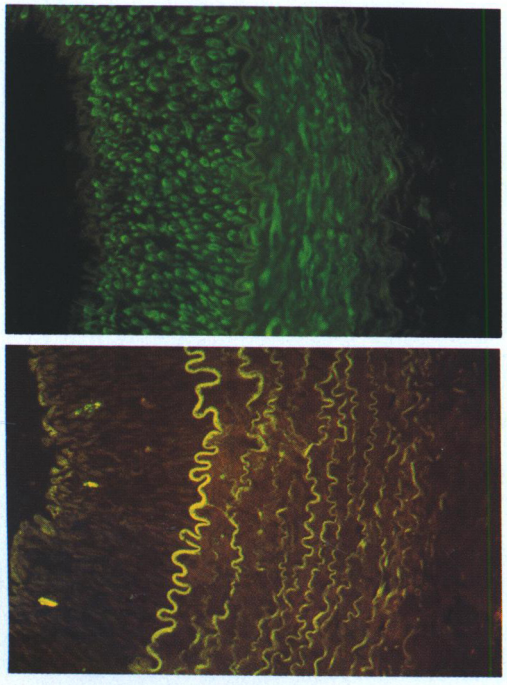


FIG 3. Carotid artery cross sections (5 μm thick) demonstrating that neointimal tissue is composed mainly of α-smooth muscle actin-1–reacting cells (smooth muscle cells) (top) with few RAM11-positive cells (macrophages) (bottom) (original magnification ×200).

Direct in vivo evaluation of cell proliferation was performed by measuring the incorporation of BrdU, a thymidine analogue, into replicating DNA. An average of 6% of total cells in control animals were BrdU-positive (Fig 4, top; Table); by double-immunofluorescence labeling techniques all BrdU-positive cells were also recognized by a specific α-SM actin antibody, further demonstrating the significant contribution of smooth muscle cell proliferation to the intimal thickening induced by perivascular manipulation.

EPC administration (groups 2 and 3) did not affect myocyte proliferation (Fig 4, middle; Table). By contrast, pretreatment with rA-$I_M$/A-$I_M$ resulted in a significant inhibition of smooth muscle cell proliferation (Fig 4, bottom; Table); the number of BrdU-positive cells was significantly reduced compared with control and EPC-treated rabbits in the media, whereas the difference in the intima was of borderline significance ($P=.065$). Again, the inhibition of smooth muscle cell proliferation was observed only in animals treated with rA-$I_M$/A-$I_M$ before collar insertion (group 4 versus group 5).

## Discussion

The present study demonstrates that a short-term treatment with intravenously delivered rA-$I_M$/A-$I_M$–containing rHDL inhibits neointimal hyperplasia caused by periarterial manipulation in cholesterol-fed rabbits. The intimal hyperplasia induced by the insertion of the extra-arterial collar results mainly from the migration and proliferation of smooth muscle cells at the site of injury. At the early stages (24 to 48 hours) of lesion development, a massive invasion of leukocytes, mainly polymorphonuclear leukocytes and monocytes, into the subendothelial space is observed (data not shown), whereas the endothelium is structurally preserved.[28,29] The infiltration of leukocytes precedes the accumulation of smooth muscle cells within the intima; within 1 week, a severe intimal thickening, consisting almost totally of proliferating smooth muscle cells, develops in normocholesterolemic animals. Such a lesion is similar to that occurring in humans after mechanical injury, eg, after endarterectomy or angioplasty.[30] Feeding animals with a cholesterol-rich diet before collar insertion results in similar proliferating lesions, which are enriched, however, by few cholesterol-loaded macrophages in the intimal layer (up to 10% of the total cells).

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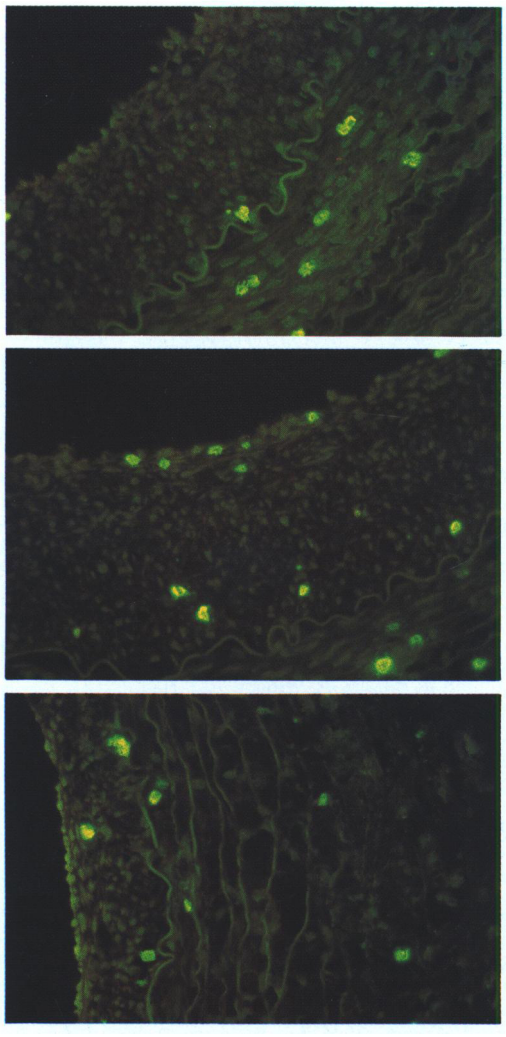


FIG 4. 5-Bromo-2′-deoxyuridine incorporation in proliferating smooth muscle cells detected by immunofluorescence. Transverse cross sections (original magnification ×400) (3 μm thick) of carotid arteries with collar from a control rabbit (top), an egg phosphatidylcholine–pretreated rabbit (middle), and a recombinant disulfide-linked apolipoprotein A-$I_{Milano}$–pretreated rabbit (bottom).

The mechanism responsible for the inhibitory effect of rA-$I_M$/A-$I_M$ on the development of intimal lesions demonstrated here remains to be clarified. In vivo and epidemiological evidence strongly suggests that HDLs exert a protective effect against atherosclerosis by promoting reverse cholesterol transport.[1] Indeed, infusions of HDL into cholesterol-fed rabbits prevented and/or removed cholesterol deposits within the arterial wall.[3] More recently, the intravenous delivery of plasma-derived human apoA-I (25 mg/wk for 6 weeks) reduced the I/M ratios in balloon catheter–injured arteries of cholesterol-fed rabbits by ≈40%.[12] In a different animal model, ie, transgenic rats overexpressing human apoA-I, modest elevations of plasma apoA-I levels similarly resulted in a significant reduction (35%) of intimal smooth muscle cell proliferation after balloon deendothelialization.[31] In spite of the obvious differences in the injury models, all these data support similar activities of apoA-I and rA-$I_M$/A-$I_M$ in inhibiting intimal thickening.

The observation of a reduced aortic cholesterol content in the apoA-I–treated animals[12] supports the concept that apoA-I, and possibly rA-$I_M$/A-$I_M$ also, inhibits intimal thickening through a stimulation of reverse cholesterol transport.[1] However, the cellular composition of the arterial lesions in the present study, as well as the short-term treatment protocol, also argues for other potential mechanisms. Since rA-$I_M$/A-$I_M$ is effective in preventing intimal thickening only when given before lesion induction, a preventive effect of rA-$I_M$/A-$I_M$ on the early events in lesion formation can be hypothesized. Indeed, apoA-I inhibits the deposition of the membrane attack complex of complement on endothelial cells,[10] an early finding preceding the appearance of overt lesions in the hypercholesterolemic rabbit model of atherosclerosis.[32] ApoA-I and rA-$I_M$/A-$I_M$ may also exert a direct effect on smooth muscle cell proliferation, resulting in a reduced incorporation of a thymidine analogue into replicating DNA. Several genes become transiently activated during cell proliferation,[33] and proto-oncogene expression is induced early in vessels after balloon denudation.[34] ApoA-I and rA-$I_M$/A-$I_M$ may affect this process, as suggested by in vitro observations of an inhibitory effect of human HDL on low-density lipoprotein–mediated stimulation of proto-oncogene expression in cultured vascular smooth muscle cells.[35]

Proliferation of vascular smooth muscle cells is one of the major mechanisms involved in the process of restenosis after successful endarterectomy or angioplasty.[36,37] The problem of restenosis after angioplasty has become critical in view of the steadily growing number of procedures and the still high rate of restenosis.[38] Many inhibitors of smooth muscle cell proliferation have been identified; some, including heparin, angiotensin-converting enzyme inhibitors, 3-hydroxy-3-methylglutaryl–coenzyme A reductase inhibitors, factor Xa inhibitors, and γ-interferon have also been found to inhibit neointimal formation in animal models of arterial restenosis.[26,39-42] However, a successful inhibitor of angioplasty restenosis in human trials has yet to be found.[43] Great effort is thus devoted to the search of innovative approaches to restenosis prevention: blockade of growth factors by monoclonal antibodies,[44] local drug delivery,[45] gene transfer,[46] and local alteration of gene expression by antisense nucleotides[47] have proven effective in animal models.

In the present study, another approach to the issue of restenosis is proposed: the use of recombinant apolipoproteins appears to drastically improve the balance between antiatherogenic and proatherogenic lipoprotein concentrations in plasma. ApoA-I seemed to be a good candidate because of its multifaceted activities on atherosclerosis-related systems[1,5-10] and because of the strong inverse correlation between plasma HDL levels and the rate of restenosis after endarterectomy[48] or angioplasty.[49] ApoA-I already proved to be effective in reducing the extent of atherosclerosis in balloon-catheterized cholesterol-fed rabbits.[12] ApoA-$I_M$/A-$I_M$, because of its improved physicochemical[20] and kinetic[19] properties versus apoA-I, might exert a more sustained effect than the normal protein, possibly reflecting the increased protection against cardiovascular disease in the A-$I_M$ carriers compared with control individuals.[16] A relative dose-response study is warranted to solve this

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issue, but such a study would require very large amounts of the wild-type and mutated proteins, which are unavailable at present. The present study was limited by the small number of animals treated in each group; therefore, the present findings should be implemented once larger amounts of protein become available. However, the very high reproducibility of intimal thickening induced by arterial manipulation and of the effects of treatment, together with the similar results obtained with apoA-I in a different animal model,[12] argue for a direct effect of apoA-I and rA-$I_M$/A-$I_M$ in preventing neointimal formation, thus supporting the potential use of recombinant apolipoproteins in the treatment of vascular disease.[13]

***Note added in proof.*** After the submission of this manuscript, we became aware of very similar results demonstrating that rA-$I_M$/A-$I_M$ reduces intimal thickening after balloon catheterization in hypercholesterolemic rabbits.[50]

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